NAME: _____ Geometry (FL B.E.S.T.)

4.4 ASA and AAS Congruence

The **ASA Postulate** proves triangles congruent from two pairs of congruent angles plus the *included* side (the side between those two angles). The **AAS Theorem** proves congruence from two pairs of congruent angles plus a non-included side — a side that lies opposite one of the given angles rather than between them. AAS actually reduces to ASA: since two angles are congruent, the Angle-Sum Theorem forces the third pair of angles to be congruent too, which makes the given side included relative to that third angle. Note that **AAA is never a valid congruence criterion** — matching all three angles only guarantees the triangles are the same shape (similar), not the same size. This worksheet applies ASA and AAS with numeric, algebraic, and overlapping-triangle reasoning, and contrasts them with AAA.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. What does ASA stand for?

2. What does AAS stand for?

3. In ASA, where must the given side be located?

4. Two angles and the included side are congruent between two triangles. Which postulate applies?

5. Two angles and a non-included side are congruent. Which postulate applies?

6. Is matching all three pairs of angles (AAA) enough to prove congruence?

7. If two pairs of angles are congruent, why must the third pair also be congruent?

8. Two triangles share an angle at a common vertex. Which property justifies the angle is congruent to itself?

9. $\triangle ABC \cong \triangle XYZ$ by ASA. $\angle B$ corresponds to which angle?

10. Marks show: angle, included side, angle. Which postulate applies?

11. Marks show: angle, angle, side opposite one of them. Which postulate applies?

12. After proving triangles congruent, corresponding sides are congruent by what idea?

13. In $\triangle ABC$ and $\triangle DEF$, $\angle A = (4x - 8)°$ and $\angle D = 60°$ are congruent; $\angle B = 45° = \angle E$; and $AB = DE = 13$ is the included side. Find x, name the postulate, then find $\angle C$ and $\angle F$.

14. Two triangular braces each have angles $50°$ and $70°$. The side opposite the $70°$ angle measures $3x - 2$ in one brace and 16 in the other, and the braces are congruent by AAS. Find x and the third angle of each brace.

15. $\overline{BD}$ bisects $\angle ABC$, with $\angle ABD = (3x + 5)°$ and $\angle CBD = (2x + 15)°$. Also $\angle A \cong \angle C = 48°$, and $\overline{BD}$ is shared (opposite the $48°$ angles). Find x, name the postulate proving $\triangle ABD \cong \triangle CBD$, and find $\angle ADB$.

✓ Answer Key

1. Angle-Side-Angle
2. Angle-Angle-Side
3. Between the two given angles (included)
4. ASA
5. AAS
6. No — AAA proves similarity, not congruence
7. By the Angle-Sum Theorem, the third angle equals $180°$ minus the other two, which are already equal
8. Reflexive Property
9. $\angle Y$
10. ASA
11. AAS
12. CPCTC
13. $x = 17$; ASA; $\angle C = \angle F = 75°$
14. $x = 6$; third angle $= 60°$ in each
15. $x = 10$; $\angle ABD = \angle CBD = 35°$; AAS; $\angle ADB = 97°$